

AUS, Thai & Nordic Bank Valuation — Stage 2

Final Report

Date: 11 July 2026 Prepared for: family review Stage: 2 complete (residual income model, trajectory-aware forecasts)

This is the closing report for Stage 2. Since the mid-project note, three things changed: (1) the universe grew from 16 to **19 banks** — DNB (Oslo), SEB (Stockholm) and Danske Bank (Copenhagen) were added, each with its own country risk build; (2) the ROE forecasting engine was rebuilt to respect each bank's actual **trajectory** instead of dragging every bank to a country average at the same speed; and (3) the model can now be run in two modes — a proper CAPM cost of capital per bank, or a flat 10% for every bank — which turns out to be a very useful way of separating “cheap because risky” from “cheap because mispriced”.

1. The model in two sentences (recap)

A bank is worth its book value plus the present value of every future dollar it earns *above* its cost of equity: residual income each year = $(\text{ROE} - \text{cost of equity}) \times \text{book value}$. We roll that forward 10 years, add a terminal value for everything beyond, divide by today's book value to get a **fair P/B**, and compare with the market's actual P/B — the gap is the mispricing signal. (The full derivation and the CBA worked example are in the mid-project report; they are unchanged.)

2. How the ROE forecasts are built now — and why each choice

The mid-project model faded every bank's ROE in a straight line to a country average over 5 years. That had a flaw we caught by inspection: it treated a bank whose ROE is *climbing* exactly like one that is *sliding*, and it let one distorted year anchor the whole forecast. The rebuilt engine works in four deliberate steps:

a) Normalise the starting point. Year-0 ROE and payout are now the **average of the last 4 reported fiscal years** plus the current trailing year — not a single snapshot. Real cases this fixed: Bank of Queensland showed a 257% payout (an earnings collapse, not generosity) and Bendigo a -1.2% ROE (one bad year in an otherwise ~7% record). Neither distortion now anchors anything.

b) Respect the trajectory. The recent trend (slope of the last ~5 annual ROEs) tilts year 1: half the observed per-year trend carries in, capped at ± 2 ROE points so four noisy data points can never launch an extreme path. Westpac (ROE rising 7.6% $\rightarrow$ 9.7%) starts its forecast *above* its average; CBA (drifting 14.7% $\rightarrow$ 12.8%) starts below. The old model treated them identically.

c) Decay toward normal at a realistic speed. From year 2, the *abnormal* part of ROE (the gap to its long-run level) shrinks by a fixed fraction each year — fast while far from normal, flattening as it lands. This matches how bank profitability actually mean-reverts, and the speed is taken from the empirical accounting literature: abnormal earnings persist at roughly **0.6–0.8 per year** (Dechow, Hutton & Sloan and successors). We use **0.80**, nudged to 0.90 for banks with a stable excess-return record (durable franchises revert slower — CBA, DNB, Bangkok Bank) and down to 0.70 for volatile records (Danske's money-laundering-era loss year, Bendigo). Every year is also hard-clipped to a 4–18% band — no bank extrapolates to impossible profitability.

d) Discipline the terminal. Beyond year 10 the model no longer assumes a country-average ROE. Instead: **terminal ROE = cost of equity + 75% of the bank's own demonstrated excess spread**, floored at -4% (chronic underperformers are not assumed to magically recover to their cost of capital —

the earlier version did exactly that, forecasting Kasikornbank to rise from 8.4% to 11.5% ROE purely because its discount rate was high, which we judged perverse and fixed). The 75% expresses that most of a proven franchise's advantage endures, but competition erodes some of it.

Why the horizon is 10 years, not 5: five years wasn't long enough for profitability to finish converging, so the terminal value was quietly capitalising a still-abnormal ROE forever. At 10 years each path visibly levels off *inside* the forecast window (Figure 1) and the terminal captures a genuine steady state.

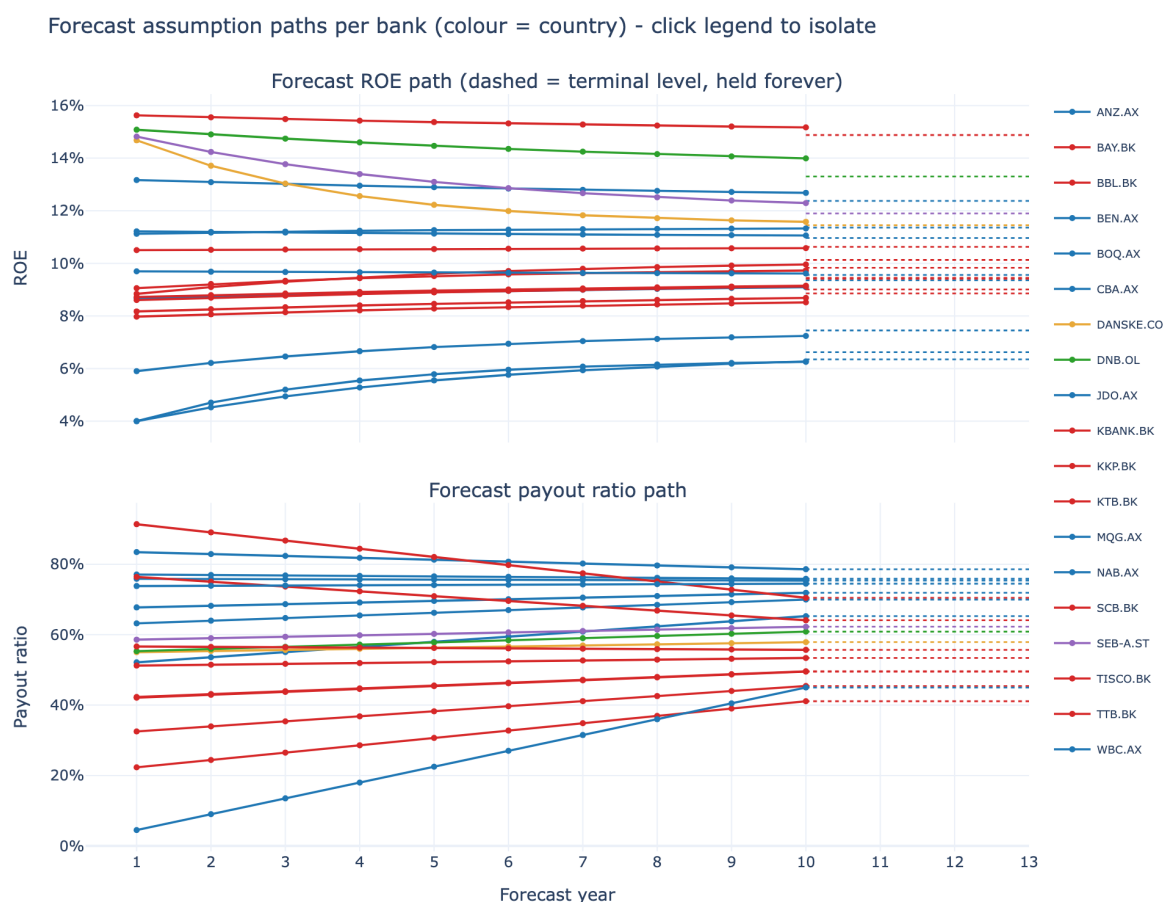


Figure 1. Every bank's forecast ROE path (top) and payout path (bottom); dashed tails = the terminal level held forever. Improvers keep momentum early, everyone glides toward a level anchored to their own demonstrated economics, and all paths level off before the terminal takes over. AUS blue, THA red, Nordics in their own colours.

Bank	Ctry	Year-1 ROE	Terminal ROE	Year-1 payout	Terminal payout
ANZ.AX	AUS	8.7%	9.4%	76%	75%
BEN.AX	AUS	4.0%	6.3%	52%	65%
BOQ.AX	AUS	4.0%	6.6%	83%	79%
CBA.AX	AUS	13.2%	12.4%	77%	76%
JDO.AX	AUS	5.9%	7.4%	4%	45%
MQG.AX	AUS	11.1%	11.4%	63%	70%
NAB.AX	AUS	11.2%	11.0%	68%	72%
WBC.AX	AUS	9.7%	9.6%	74%	74%
DANSKE.CO	DNK	14.7%	11.5%	55%	58%
DNB.OL	NOR	15.1%	13.3%	55%	61%

Bank	Ctry	Year-1 ROE	Terminal ROE	Year-1 payout	Terminal payout
SEB-A.ST	SWE	14.8%	11.9%	59%	62%
BAY.BK	THA	8.0%	8.9%	22%	41%
BBL.BK	THA	8.2%	9.0%	33%	45%
KBANK.BK	THA	8.6%	9.4%	42%	50%
KKP.BK	THA	8.8%	10.1%	51%	53%
KTB.BK	THA	10.5%	10.6%	42%	50%
SCB.BK	THA	9.1%	9.8%	76%	64%
TISCO.BK	THA	15.6%	14.9%	91%	70%
TTB.BK	THA	8.7%	9.4%	57%	56%

Table 1. Where each path in Figure 1 starts (year 1) and where it ends up (terminal, held forever) - the chart's start and end points in numbers. Generated from the same model run as the chart.

3. What the market pays for: ROE vs P/B

Before any modelling, one chart explains most of bank pricing: the market pays higher price-to-book for higher return-on-equity. The valuation question is never *whether* CBA deserves a premium to Bangkok Bank — it clearly does — but whether the premium is the *right size* given how long that profitability can last and the return shareholders require. That is exactly what the residual income model prices.

ROE vs actual P/B

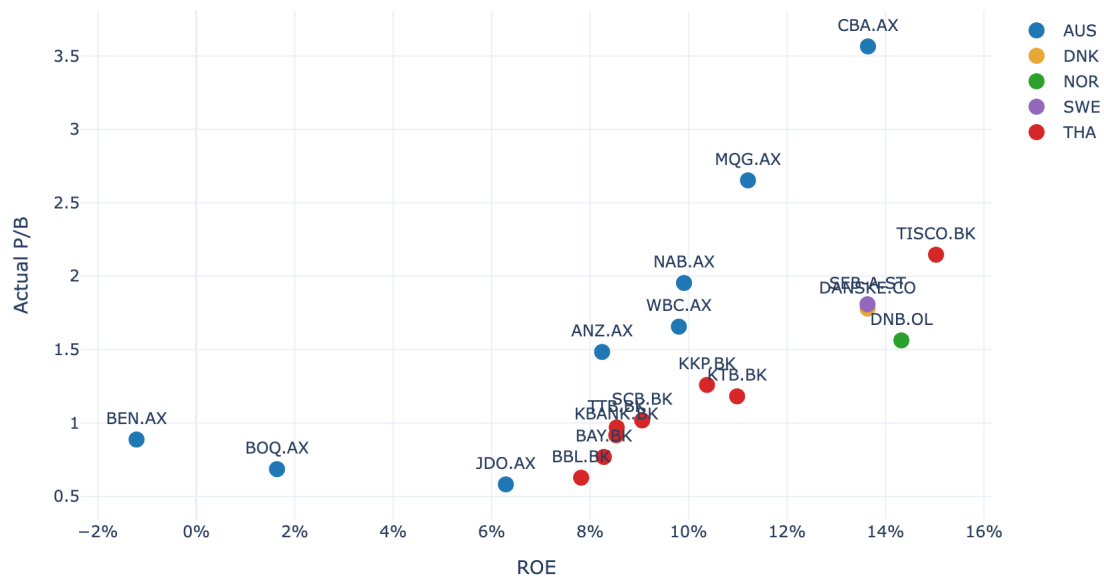


Figure 2. Current ROE vs actual market P/B. The upward sweep is the market paying for profitability; banks far above the sweep (CBA at 3.6× book) are priced for that profitability lasting a very long time.

4. Cost of capital per bank (CAPM — abbreviated recap)

Same construction as the mid-project report (risk-free rate + beta × equity premium + country premium; Thailand carries a +3% country premium for FX/political/liquidity risk, applied as a parallel spread across all Thai banks). Two additions since then. **Nordics:** Norway, Sweden and Denmark enter as AAA markets with no country premium and their own (low) bond yields. **A beta judgment call:** the measured Nordic betas (0.34–0.59 against local indices) produced costs of equity near 4.5–5.5% — implausible for leveraged banks and an artifact of index composition — so all three are floored at 0.90. A leveraged bank cannot credibly be half as risky as its home market.

Bank	Ctry	Beta	Risk-free	Equity prem.	Country prem.	Cost of equity
CBA	AUS	0.95	4.3%	5.5%	0.0%	9.5%
WBC / NAB	AUS	1.05	4.3%	5.5%	0.0%	10.1%
ANZ / BEN	AUS	1.10	4.3%	5.5%	0.0%	10.4%
MQG	AUS	1.25	4.3%	5.5%	0.0%	11.2%
BOQ	AUS	1.15	4.3%	5.5%	0.0%	10.6%
JDO	AUS	1.30	4.3%	5.5%	0.0%	11.5%
BBL / SCB / TTB	THA	0.95	2.5%	7.0%	3.0%	12.2%
KBANK	THA	1.00	2.5%	7.0%	3.0%	12.5%
KTB / TISCO	THA	0.90	2.5%	7.0%	3.0%	11.8%
BAY	THA	0.80	2.5%	7.0%	3.0%	11.1%
KKP	THA	1.10	2.5%	7.0%	3.0%	13.2%
DNB	NOR	0.90 (floored)	3.8%	5.0%	0.0%	8.3%
SEB	SWE	0.90 (floored)	2.4%	5.0%	0.0%	6.9%
Danske	DNK	0.90 (floored)	2.3%	5.0%	0.0%	6.8%

Table 2. CAPM cost of equity for all 19 banks (grouped where identical). Note the consequence of low Nordic bond yields: even with the beta floor, SEB and Danske carry required returns under 7% — the lowest hurdle in the universe.

5. The experiment: CAPM cost of capital vs a flat 10% for everyone

Every valuation above depends on the required return, and the required return is itself an estimate. So we ran the entire model twice: once with the per-bank CAPM numbers (Table 2), and once forcing **every bank to exactly 10%**. The flat run strips out all differences in discount rate — what remains is a pure profitability-versus-price comparison. Reading the two together tells you *why* each bank screens cheap or expensive:

- Cheap under CAPM but not at 10% → the appeal comes from a *low discount rate*, not superior economics.
- Cheap in **both** → robust — the bank out-earns even a demanding hurdle at its current price.
- Expensive in both → the market is paying for more than the demonstrated economics under any reasonable required return.

Bank	Actual P/B	Fair P/B (CAPM)	Mispricing (CAPM)	Fair P/B (10%)	Mispricing (10%)	Read
DNB	1.56	2.37	+52%	1.77	+13%	Robust: cheap in both
SEB	1.81	2.66	+47%	1.68	−7%	Mostly a low-rate effect
Danske	1.78	2.59	+46%	1.46	−18%	Mostly a low-rate effect
BBL	0.63	0.54	−13%	0.71	+13%	Cheap once risk equalised
BAY	0.77	0.60	−22%	0.70	−9%	Near fair once risk equalised
SCB	1.02	0.71	−30%	0.89	−13%	Part risk premium, part rich
KTB	1.18	0.83	−30%	1.05	−11%	Part risk premium, part rich
JDO	0.58	0.39	−34%	0.38	−35%	Expensive in both
TTB	0.97	0.64	−34%	0.81	−17%	Part risk premium, part rich
TISCO	2.15	1.40	−35%	1.73	−20%	Expensive in both
BOQ	0.69	0.44	−36%	0.42	−39%	Expensive in both
KBANK	0.92	0.58	−36%	0.78	−15%	Part risk premium, part rich
NAB	1.95	1.15	−41%	1.16	−41%	Expensive in both
ANZ	1.48	0.84	−44%	0.87	−41%	Expensive in both
WBC	1.66	0.93	−44%	0.94	−43%	Expensive in both
KKP	1.26	0.61	−51%	0.87	−31%	Expensive even risk-equalised
BEN	0.89	0.40	−54%	0.39	−56%	Expensive in both
CBA	3.56	1.50	−58%	1.40	−61%	Expensive in both
MQG	2.65	1.02	−62%	1.17	−56%	Expensive in both

Table 3. All 19 banks under both modes, ranked most- to least-attractive under CAPM. The Nordic story splits cleanly: DNB survives a 10% hurdle on sheer profitability; SEB and Danske's apparent cheapness is largely their sub-7% CAPM discount rate. "Part risk premium, part rich" = the gap roughly halves when Thai risk is equalised, but a discount remains.

Mispricing by bank (fair P/B / actual P/B - 1). Positive = undervalued

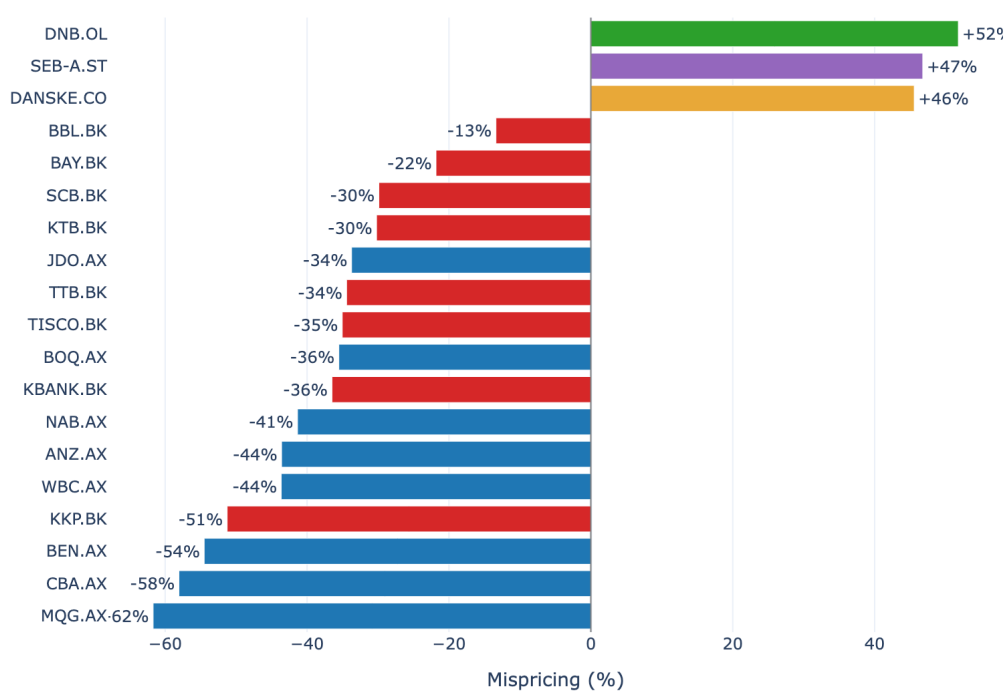


Figure 3a. Mispricing ranking under CAPM cost of capital. Positive = undervalued.

Mispricing by bank (fair P/B / actual P/B - 1). Positive = undervalued

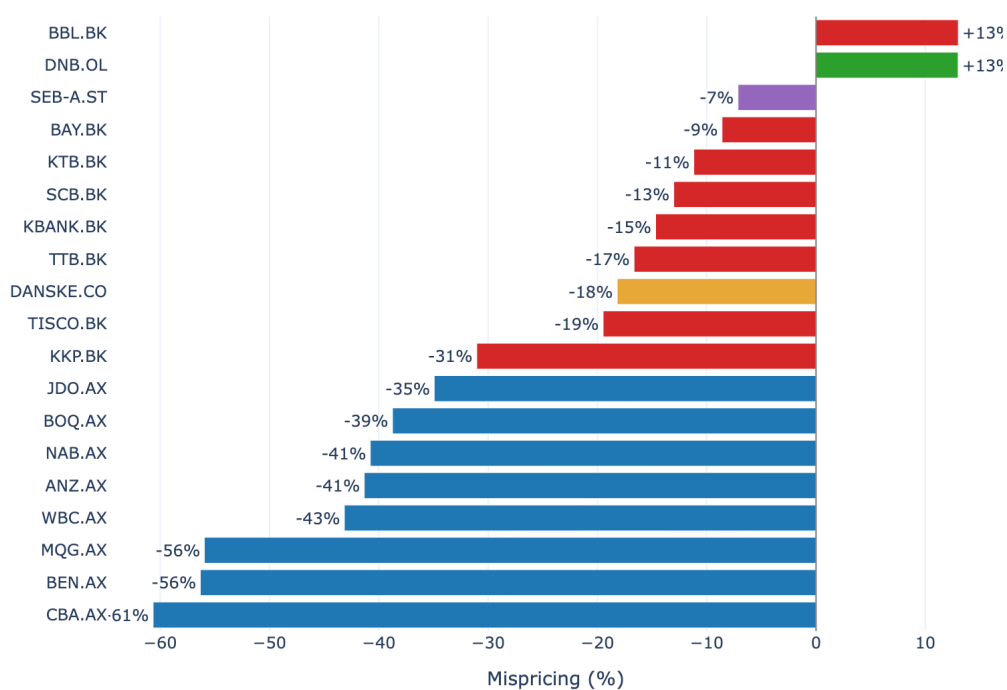


Figure 3b. The same ranking with every bank at a flat 10% cost of capital. The Nordics compress sharply (their low discount rate was most of the story), Thai banks improve (their 12–13% risk penalty is removed), and the Australian majors stay at the bottom in both worlds.

Fair P/B by scenario (whiskers = bear..bull) vs current market P/B

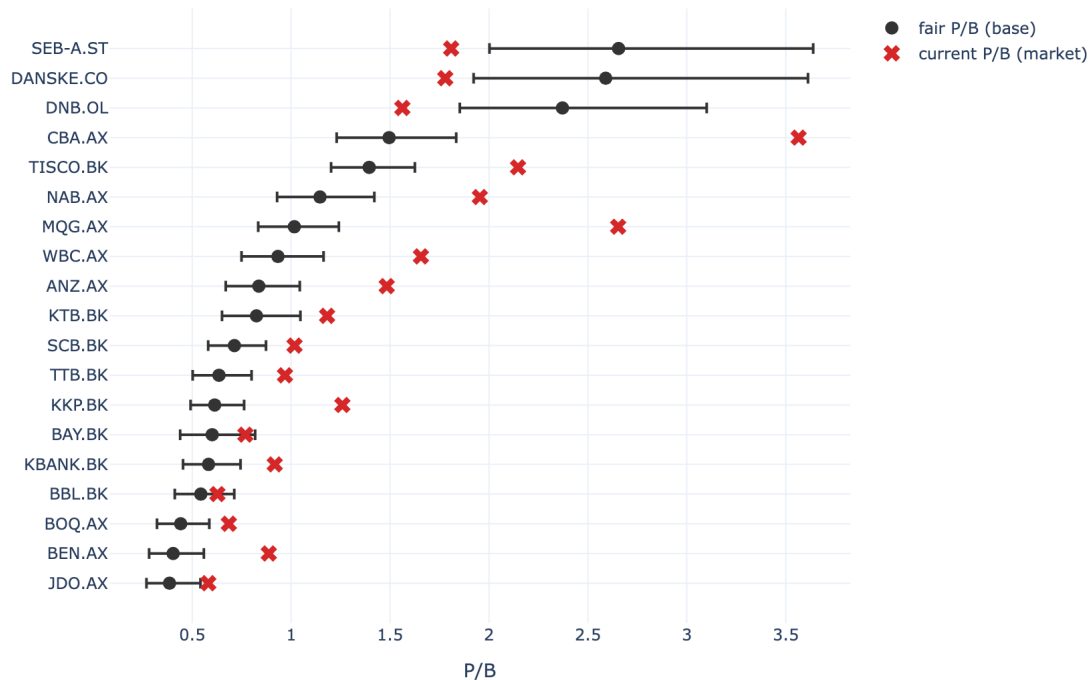


Figure 4a. Fair P/B under bear/base/bull scenarios (whiskers) vs current market P/B (red x), CAPM mode. Where the x sits beyond the bull whisker, the market is paying above even the optimistic case.

Fair P/B by scenario (whiskers = bear..bull) vs current market P/B

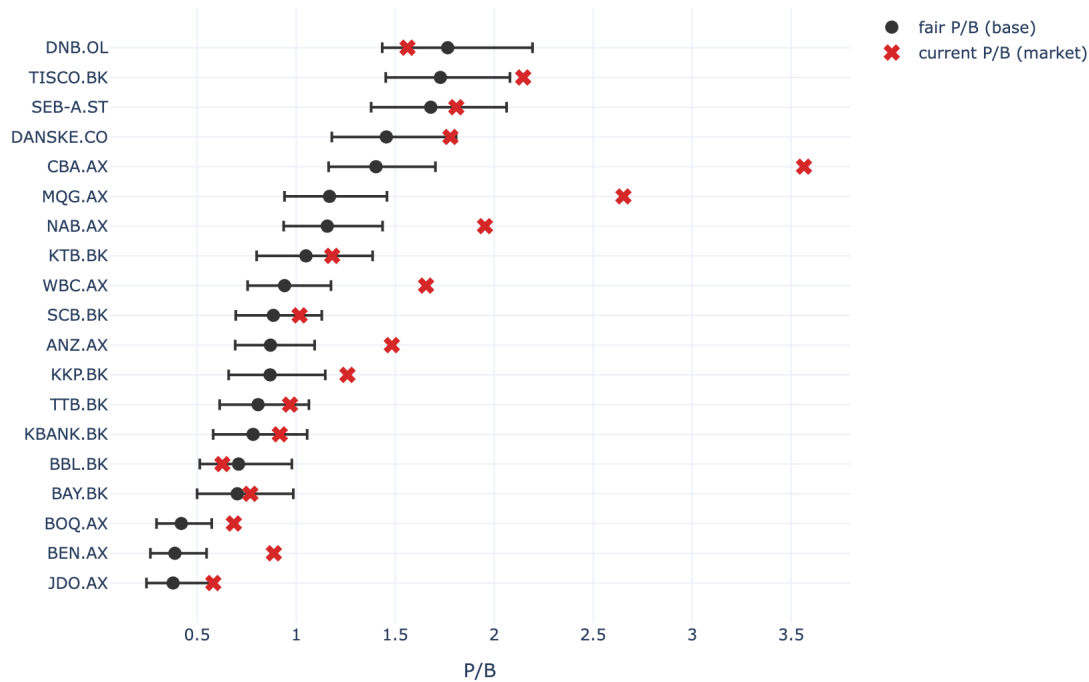


Figure 4b. The same scenario view at a flat 10% cost of capital. CBA and Macquarie sit far beyond their bull whiskers in *both* modes — that conclusion does not depend on the discount-rate assumption.

6. Current headline assumptions (all editable in one file)

Assumption	Value	Why this value
Forecast horizon	10 years	Long enough for ROE to finish converging before the terminal
History used	4 years + current	Averages out one-off distortions; reads the trend
Trend carried into year 1	50%, max ± 2 pts	Momentum respected, noise capped
Persistence of abnormal ROE	0.80 (0.70–0.90)	Empirical range 0.6–0.8; stable franchises at the top end
ROE hard band	4% – 18%	No bank extrapolates beyond real-world bank profitability
Franchise spread kept forever	75%	Most of a proven moat endures; competition erodes the rest
Terminal spread floor	–4%	Weak banks stay weak; no assumed recovery to cost of capital
Long-run growth	AUS 3.5% / THA 4.0% / Nordics 3.0–3.5%	Nominal GDP-like growth of the perpetuity

Table 4. Key settings in config/assumptions.yaml. Every number above is one edit away from a rerun — nothing is hard-wired.

Where things stand. The model now starts from normalised multi-year profitability, respects each bank's direction of travel, reverts at an empirically grounded speed, and refuses to assume either that winners keep everything or that losers recover. Run under two cost-of-capital regimes, the conclusions that survive both are: **DNB looks genuinely cheap; the Australian majors (CBA, Macquarie, NAB) are priced beyond even bull-case fair value; Thai banks are roughly fair once their risk premium is the explanation rather than the verdict; SEB and Danske are cheap only if you accept a sub-7% required return.** Honest caveats: Nordic profitability is measured at a rate-cycle peak; SCB's history is unavailable from the data source (snapshot-only forecast); and none of this is backtested yet — Stage 4 will test whether the fair-P/B signal actually predicts returns before any money moves.